Fruit Market · Decision register

Generated by `npm run records` from `docs/workflow/records/`. Do not edit by hand. Last recorded change: 2026-10-26 15:34 (Asia/Kuala_Lumpur).

Approval status and delivery status are separate. Only the owner's explicit words set "approved".

Pending approvals

ID	What	Status
DEC-001	Adopt the personal workflow guide 1.2 for Fruit Market	Proposed
DEC-004	PDF exports with pandoc and the existing Playwright browser	Proposed
DEC-005	Budget interpretation: RM500 per month across all games	Proposed
PLAN-003	Package C: online, capacity and release evidence (outline only)	Proposed

Current decisions

ID	Decision	Approval	Delivery	Reports
DEC-001	Adopt the personal workflow guide 1.2 for Fruit Market	Proposed	In progress	REP-001
DEC-002	Package A scope = PLAN-001	Approved (owner, 2026-10-26)	Verified	REP-001, REP-002
DEC-003	Live deploys only through the manual Deploy workflow after verify	Approved (owner, 2026-10-26)	Verified	REP-001, REP-002
DEC-004	PDF exports with pandoc and the existing Playwright browser	Proposed	Implemented	REP-001
DEC-005	Budget interpretation: RM500 per month across all games	Proposed	Not started	REP-001
DEC-006	Preview = separate staging Worker	Approved (owner, 2026-10-26)	Verified	REP-002, REP-003
DEC-007	Package B scope = PLAN-002	Approved (owner, 2026-10-26)	Verified	REP-003, REP-004
DEC-008	Canonical record format: JSON records plus Markdown narrative reports	Approved (owner, 2026-10-26)	Verified	REP-003, REP-004
DEC-009	Records check runs in verify and CI	Approved (owner, 2026-10-26)	Verified	REP-003, REP-004
DEC-010	Reconnect fix scope = PLAN-004 (evidence first, then fix)	Approved (owner, 2026-10-26)	Verified	REP-005, REP-006
DEC-011	Stress test runs on demand, not on every push	Approved (owner, 2026-10-26)	Verified	REP-005, REP-006

Work packages

ID	Package	Approval	Delivery	Release
PLAN-001	Package A: baseline, one verification path, preview and owner brief	Approved (owner, 2026-10-26)	Verified	Unreleased
PLAN-002	Package B: canonical records, generated views and checks	Approved (owner, 2026-10-26)	Verified	Unreleased
PLAN-003	Package C: online, capacity and release evidence (outline only)	Proposed	Not started	Unreleased
PLAN-004	Reconnect fix: the game screen sometimes doesn't return after rejoining an online room	Approved (owner, 2026-10-26)	Verified	Unreleased

Decision details

DEC-001 · Adopt the personal workflow guide 1.2 for Fruit Market

- **Question:** Should Fruit Market be run with the workflow guide (one approved package at a time, evidence-based checks, decision records)?
- **Choice:** Yes; one package at a time, starting with Package A.
- **Rationale:** The game is complete but releases were unreviewed and records lived in chat history.
- **Player impact:** None directly; fewer broken releases.
- **Technical impact:** Adds checks, records and a release gate.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Proposed
- **Delivery:** In progress
- **Review trigger:** If the process costs more time than it saves after three packages.
- **Reports:** REP-001

DEC-002 · Package A scope = PLAN-001

- **Question:** What should the first setup package contain?
- **Choice:** One verify command, preview before publishing, short records and an owner brief (PLAN-001).
- **Rationale:** Smallest package that stops unreviewed releases and gives the owner a clear way to try builds.
- **Player impact:** None until a release; protects players from silent breakage.
- **Technical impact:** package.json scripts, CI and deploy workflows, new docs.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-001 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** Acceptance A4 (owner preview playtest) still pending.
- **Reports:** REP-001, REP-002

DEC-003 · Live deploys only through the manual Deploy workflow after verify

- **Question:** When should the live game change?
- **Choice:** Only when the owner runs Deploy LIVE game and types PUBLISH; it runs verify first and a smoke test after. Never on push.
- **Rationale:** Makes 'a preview is not a release' true for this repository.
- **Player impact:** Updates arrive only after the owner approves them.
- **Technical impact:** deploy-web.yml is manual only.
- **Data and security:** None.
- **Cost:** RM0; one extra tap per release.
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-001 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** If urgent fixes are delayed because nobody pressed Deploy.
- **Reports:** REP-001, REP-002

DEC-004 · PDF exports with pandoc and the existing Playwright browser

- **Question:** Which tool builds the decision PDFs?
- **Choice:** pandoc (Markdown to HTML) plus the project's existing Playwright Chromium (HTML to PDF). No installs.
- **Rationale:** Both are already available; no new dependency.
- **Player impact:** None.
- **Technical impact:** PDF export runs locally, not in CI.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Proposed
- **Delivery:** Implemented
- **Review trigger:** If pandoc is unavailable on the machine that exports.
- **Reports:** REP-001

DEC-005 · Budget interpretation: RM500 per month across all games

- **Question:** How should the RM500 monthly ceiling be applied?
- **Choice:** Treat it as portfolio-wide. Fruit Market's current spend is unknown (hypothesis: Cloudflare Free, or Workers Paid at about USD 5).
- **Rationale:** The workflow guide's conservative default.
- **Player impact:** None.
- **Technical impact:** Paid services need a quote and approval.
- **Data and security:** None.
- **Cost:** Unknown; to be recorded.
- **Owner:** Murshidi
- **Approval:** Proposed
- **Delivery:** Not started
- **Review trigger:** When the owner states the Cloudflare plan and other paid accounts.
- **Reports:** REP-001

DEC-006 · Preview = separate staging Worker

- **Question:** How should changes be previewed, given that Cloudflare preview and version URLs are not generated for Workers with Durable Objects?
- **Choice:** A separate Worker, fruit-market-staging, with its own rooms, leaderboard and stats, updated by CI after checks pass on main.
- **Rationale:** Closest free option to a per-change preview URL; keeps the live data untouched.
- **Player impact:** None; players never see the preview.
- **Technical impact:** A second Worker and a preview job in CI.
- **Data and security:** Separate storage; preview link is unlisted, not private.
- **Cost:** No new paid service; tiny usage within the existing plan.
- **Owner:** Murshidi
- **Approval:** Approved: “Confirm DEC-006, and plan PLAN-002” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** If Cloudflare adds preview URLs for Durable Object Workers.
- **Reports:** REP-002, REP-003

DEC-007 · Package B scope = PLAN-002

- **Question:** What should the records package contain?
- **Choice:** Canonical JSON records, generated views (register, changelog, state block, PDFs), a records check in verify, release tagging on Deploy, and migration of existing records.
- **Rationale:** Records were kept by hand in several places with nothing checking that they agreed.
- **Player impact:** None directly; releases become traceable.
- **Technical impact:** New records, a generator and checker script, and a deploy step.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)
- **Review trigger:** If keeping records costs more than it saves.
- **Reports:** REP-003, REP-004

DEC-008 · Canonical record format: JSON records plus Markdown narrative reports

- **Question:** Where is the single source of truth for decisions and packages?
- **Choice:** JSON files in docs/workflow/records; hand-written REP reports stay as narrative snapshots; Markdown, changelog and PDF views are generated.
- **Rationale:** Node reads JSON without dependencies and the check can be strict. YAML needs a parser; plain Markdown can't be validated; a database is ruled out by the guide.
- **Player impact:** None.
- **Technical impact:** Claude edits the JSON; the owner reads the generated views.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi

- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)
- **Review trigger:** If hand edits to JSON become a frequent source of errors.
- **Reports:** REP-003, REP-004

DEC-009 · Records check runs in verify and CI

- **Question:** Should record disagreements fail the build?
- **Choice:** Yes: npm run verify runs records:check, so CI fails on broken records.
- **Rationale:** Once branch protection is on, broken records can't be merged.
- **Player impact:** None.
- **Technical impact:** verify is about 1 second slower.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)
- **Review trigger:** If the check blocks urgent fixes for record reasons.
- **Reports:** REP-003, REP-004

DEC-010 · Reconnect fix scope = PLAN-004 (evidence first, then fix)

- **Question:** How should the intermittent rejoin failure be fixed?
- **Choice:** Collect evidence and reproduce it with a stress test before changing game code; fix only the measured root cause, with a regression test.
- **Rationale:** Reading the code didn't identify a certain cause, and two failures in five runs are too rare to debug by guessing. The workflow guide says to reproduce and isolate before fixing.
- **Player impact:** Fewer stuck screens after a reload or dropped connection, once the fix is published.
- **Technical impact:** New stress test and diagnostics; a small fix in the online client or room server.
- **Data and security:** Diagnostics stay in CI artifacts for 7 days; they contain test players only.
- **Cost:** RM0 (GitHub Actions minutes are free for public repositories)
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-004 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-004)
- **Review trigger:** If not reproduced after 300 cycles: stop and report.
- **Reports:** REP-005, REP-006

DEC-011 · Stress test runs on demand, not on every push

- **Question:** Should the long reconnect stress test run in every CI run?
- **Choice:** No: a manual CI workflow plus local runs. Every-push CI keeps the existing tests (about 15 minutes).

- **Rationale:** Hundreds of cycles would make every CI run far slower; the stress test is for investigation and before releases.
- **Player impact:** None.
- **Technical impact:** One extra manual workflow.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-004 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-004)
- **Review trigger:** If reconnect bugs slip through between releases.
- **Reports:** REP-005, REP-006

History (oldest first)

- 2026-10-26 13:15 · REP-001: baseline audit; PLAN-001 proposed; DEC-001 to DEC-005 proposed.
- 2026-10-26 13:20 · Owner: "Approve PLAN-001 only, including its listed tests and documentation". DEC-002 and DEC-003 approved; PLAN-001 approved.
- 2026-10-26 13:25 · PLAN-001 implemented on main (2e4b240). Push started CI and Android only; no live deploy.
- 2026-10-26 13:42 · REP-002: PLAN-001 verified except A4 (owner playtest). DEC-006 recorded, awaiting confirmation.
- 2026-10-26 13:46 · Owner: "Confirm DEC-006, and plan PLAN-002". DEC-006 approved. REP-003: PLAN-002 proposed; DEC-007 to DEC-009 awaiting approval.
- 2026-10-26 13:48 · Owner: "Approve PLAN-002 only, including its listed tests and documentation". PLAN-002, DEC-007, DEC-008 and DEC-009 approved.
- 2026-10-26 13:50 · Correction: the report headers of REP-002 (13:45) and REP-003 (13:50) were rounded estimates; the commit times 13:42 and 13:46 are used above. The reports are left as written.
- 2026-10-26 13:52 · PLAN-002 implemented on workflow/plan-002: records migrated; B1, B2, B3, B5 and B7 PASS; B4 and B6 pending.
- 2026-10-26 14:09 · PLAN-002 verified: B1 to B7 PASS (CI run 40276699547 on a7fa577). DEC-007, DEC-008 and DEC-009 delivered.
- 2026-10-26 14:09 · Known issue found: CI run 40275680793 (docs-only commit 63c3682) failed because the online test's page reload did not show the game within 20 s. Same symptom as the quick-match failure in CI run 1. Pre-existing and intermittent (2 of 5 CI runs); not part of PLAN-002. Proposed as the next package in REP-004.
- 2026-10-26 14:16 · Owner: "PLAN the reconnect fix". REP-005: PLAN-004 proposed; DEC-010 and DEC-011 awaiting approval.
- 2026-10-26 14:18 · Owner: "Approve PLAN-004 only, including its listed tests and documentation". PLAN-004, DEC-010 and DEC-011 approved. (Time approximate: between the REP-005 commit at 14:17 and the first work commit at 14:27.)
- 2026-10-26 14:53 · PLAN-004: root cause reproduced (lost first update after rejoining; 20 of 20 hidden-tab reloads stuck before the fix) and fixed in the client (9d9e911).
- 2026-10-26 15:01 · PLAN-004: R1 to R5 PASS (165 hidden-tab cycles after the fix, 0 failed). R6 waiting for 3 CI runs.

- 2026-10-26 15:34 · REP-006: PLAN-004 verified (R1 to R6 PASS). The fix is on the preview; the live game is unchanged until a release is prepared and deployed.

REP-001 · Baseline audit and PLAN-001 (Package A)

Game	Fruit Market (`murshidihashim/Fruit-Market`)
Report date	2026-10-26 13:15 (Asia/Kuala_Lumpur)
Build audited	`main @ c915e49` (CI run 2 green, deployed)
Prepared by	Claude (studio lead), from the "Personal workflow for Murshidi", edition 1.2
Type	Planning only. **No game implementation changes were made.**
Approval status	PLAN-001 **awaiting approval**

1 · Baseline: where the game stands

Stage: between *MVP / private alpha* and *browser beta*. The game is complete and publicly reachable, but no unfamiliar players have been observed yet, so player understanding and fun are unproven.

Working and must be preserved

- Rules engine: deterministic, seeded, a single entry point (`applyAction`), rules version 2. 63 unit tests, including 200 simulated games.
- Saves: versioned envelope (`v2` plus rules version); old saves still load.
- Online play is server-authoritative (Cloudflare Worker with Durable Objects):
  - secret prices are redacted; the leak check reads every network message and found 0 leaks
  - reconnect, a 75 s grace period then computer takeover, and quick match
  - per-IP rate limits, a name policy, and a replay-verified daily leaderboard
- Android build (Capacitor) is produced by CI on every push.
- Docs: `RULES.md` , `ONLINE.md` , `PUBLISHING.md` , `docs/ANALYTICS.md` , the studio audit `docs/studio/01–10` (backlog = `10`).

Gaps against the workflow guide

#	Finding	Type	Severity
G1	Every push to `main` deploys straight to the public URL. The guide requires a separate publication approval.	Process defect	High
G2	`main` has **no branch protection** (checked via GitHub: `protected: false`). CI results don't block anything.	Measured	High
G3	No single verification command. There are separate `lint` , `typecheck` , `test` , `build` , `test:e2e` scripts.	Measured	Medium
G4	No `CLAUDE.md` , current-state record, game brief, or owner brief. Each session relies on long chat history.	Measured	Medium
G5	No decision records or PDF register before this report.	Measured	Low
G6	Real-device testing (Android Chrome, iPhone Safari): never done.	NOT RUN	High for release

#	Finding	Type	Severity
G7	No backup or restore for server data (leaderboards, stats). Room crash recovery not tested.	NOT RUN	Medium
G8	Monthly hosting cost unknown (Cloudflare plan not recorded).	Unknown	Medium
G9	The game-screen open time is noisy (217–489 ms locally). Quick match failed once in CI and passed on the rerun.	Measured, pre-existing	Low

2 · Readiness baseline (evidence as of today)

Gate	Check	Result	Evidence and limitation
Build ready	Clean install, lint, typecheck, unit tests, build	PASS	CI run 40266399588 on `c915e49`
Play ready	Automated solo game to results	PASS	`scripts/smoke.cjs` , local and CI
Play ready	Owner completes a match in a preview	NOT RUN	Needs you
Online ready	Two independent clients finish a match; results agree	PASS	`e2e-online.cjs` , local and CI run 2
Online ready	Private state: no secret prices in any message	PASS	40 frames checked, 0 leaks
Online ready	Illegal actions, wrong phase, votes, takeover	PASS	`tests/room.test.ts` (16 tests)
Online ready	Reconnect, abandonment, quick match	PASS	e2e; one CI flake, then a pass
Online ready	Server crash recovery	NOT RUN	Room state is persisted per action; never tested
Release ready	Real Android Chrome and iPhone Safari	NOT RUN	Emulation only
Release ready	Backup and restore, rollback drill	NOT RUN	—
Release ready	Live smoke test after deploy	NOT RUN	The sandbox proxy blocks the live URL; the deploy job itself succeeds
Release ready	Publication approval gate	FAIL	See G1

Release-ready is **not claimed**.

3 · PLAN-001 · Package A: baseline, one verification path, preview and owner brief

Player problem: updates reach players without your approval and without one clear pass/fail check, so a bad change could ship silently. You also have no short guide for trying a build yourself.

Included scope

1. **One verification command.** `npm run verify` runs lint, typecheck, unit tests and build, and stops at the first failure (about 1 minute). `npm run verify:full` adds the browser suite (about 15 minutes). CI calls `verify:full`. Existing scripts are reused; nothing new is installed.
2. **Preview before publication.**
   - Pushes to `main` run checks only.
   - Production deploys happen only when you trigger *Deploy* (a manual GitHub button), which first runs `verify`.
   - Each change also gets a private Cloudflare **preview URL** that doesn't touch the live game. I'll confirm in Cloudflare's current docs that this is free on your plan; if it isn't, the local preview steps are the fallback.
3. **Branch protection instructions.** A 5-step guide for you to turn it on in GitHub settings (I can't change repository settings). It requires the CI check before merging.
4. **Short records** (reusing the studio docs, not duplicating them):
   - `CLAUDE.md`, about 90 lines: rules, actual commands, preservation constraints, pointers
   - `docs/workflow/STATE.md`, about 500 words
   - `docs/workflow/GAME-BRIEF.md`: audience hypothesis, core loop, session length, exclusions
5. **Owner brief.** `docs/workflow/OWNER-BRIEF.md`: how to open a preview, a 10-minute playtest checklist, and what each check means in plain English.

Excluded: any change to rules, balance, art, saves, the server protocol or hosting; paid services; publication; Package B (record automation) and Package C (devices, backups, capacity).

Roles: producer (scope), DevOps and release (deploy gate, CI), QA (verify command), technical lead (`CLAUDE.md`). Self-review only; low risk.

Likely files: `package.json` (scripts), `.github/workflows/{ci,deploy-web}.yml`, new `CLAUDE.md`, new `docs/workflow/{STATE,GAME-BRIEF,OWNER-BRIEF}.md`, `README.md` (pointer).

Acceptance (observable)

- A1. `npm run verify` exits 0 on the current code. A deliberately broken test makes it exit non-zero (shown, then reverted).
- A2. Pushing to `main` does **not** change the live site. Running *Deploy* does, and only after `verify` passes.
- A3. A preview URL (or documented local preview) lets you play the build before you approve publication.
- A4. You can follow `OWNER-BRIEF.md` to open a preview and finish a match, without asking me anything.
- A5. `CLAUDE.md` stays under 120 lines; `STATE.md` stays at or under about 600 words.

Checks: `verify` (PASS and FAIL demonstration), one CI run, one manual deploy dry run (after your separate publication approval, or skipped and marked NOT RUN), and the line and word counts.

Cost and data: RM0 extra. No player data is touched.

Risks: a preview URL feature may need a Cloudflare setting (fallback: local preview). If you forget to press *Deploy*, fixes wait (that's the point).

Rollback: revert the package commit; the deploy workflow returns to deploying on every push.

4 · Packages B and C (outline only; detailed later)

- **PLAN-002 · Package B, records and PDFs:**
  - one canonical record per package (Markdown with an ID header)
  - a deterministic local script that builds the owner brief, changelog, report PDFs and cumulative register, using pandoc and the project's existing Playwright browser (no new dependency)
  - export failures are labelled PDF EXPORT BLOCKED
- **PLAN-003 · Package C, online, capacity and release evidence:**
  - real-device checklist runs (you plus your phones)
  - Durable Object data export and restore drill
  - a room crash-recovery test
  - a small load test within an approved budget, with written workload assumptions (peak concurrent players, 2–4 per room, about 15 min matches)
  - cost record with a dated Cloudflare quote
  - live smoke test from an unrestricted machine (CI)

5 · Decisions

ID	Question	Recommendation	Status	Delivery
DEC-001	Adopt the workflow guide 1.2 for Fruit Market?	Yes; one package at a time	Proposed	Not started
DEC-002	What goes in Package A?	PLAN-001 scope above	Awaiting approval	Not started
DEC-003	When does production deploy?	Only on your manual *Deploy* after `verify`; never on push	Awaiting approval	Not started
DEC-004	PDF tooling	pandoc plus the existing Playwright browser; no installs	Proposed	Used for this report
DEC-005	Budget interpretation	RM500/month portfolio-wide ceiling; current Fruit Market spend unknown (hypothesis: Cloudflare Free, or Workers Paid at about USD 5)	Proposed (assumption)	Not started

Rationale for DEC-003: it is the only way to make "a preview is not a release" true for this repo. Trade-off: one extra tap per release.

6 · Questions for you (max 3)

1. **Devices:** which real phones can you test on (Android model, iPhone model)? This decides Package C.
2. **Hosting cost:** is your Cloudflare account on the Free plan or Workers Paid? Any other paid accounts yet (domain, Google Play developer account)?
3. **Approve PLAN-001** as written (including DEC-003)? Or change it?

7 · Sources

- Repository inspection, `main @ c915e49`, 2026-10-26.
- GitHub branch API (`main protected: false`), 2026-10-26.
- CI run 40266399588 (green), deploy run 40266399518 (success), Android run 40266399555 (success).
- Workflow guide: "Personal workflow for Murshidi", consolidated edition 1.2, 26 Oct 2026.
- Preview-URL and plan-pricing facts to be confirmed against current Cloudflare docs during PLAN-001 (not yet verified).

REP-002 · PLAN-001 (Package A) implemented and verified

Game	Fruit Market
Report date	2026-10-26 13:45 (Asia/Kuala_Lumpur)
Package	PLAN-001, approved by the owner 2026-10-26: "Approve PLAN-001 only, including its listed tests and documentation"
Build	`main @ 2e4b240` ; records on `workflow/plan-001`
Live game	**Unchanged** (still `c915e49`). No publication approval was given and none was performed.
Delivery status	Implemented; verified except A4, which needs the owner

What changed

Added

- `npm run verify` : lint, types, unit tests, build; stops at the first failure.
- `npm run verify:full` : `verify` plus browser tests.
- A **preview** copy of the game, updated automatically after CI passes on `main` , then smoke-tested: https://fruit-market-staging.murshidi-hashim.workers.dev. It is a separate Worker with its own rooms, leaderboard and stats.
- `scripts/smoke-live.mjs` : after a deploy, checks that the page loads and the server runs the expected rules version.
- `CLAUDE.md` (72 lines), `docs/workflow/STATE.md` , `GAME-BRIEF.md` , `OWNER-BRIEF.md` (preview, playtest checklist, publishing, branch-protection steps).

Changed

- The live deploy now runs only through GitHub Actions → *Deploy LIVE game* → type `PUBLISH` . It runs `verify` first and a live smoke test after.
- CI runs `verify:full` . README and ONLINE.md describe the new path.
- Lint now covers the docs export script (it found 12 errors in it, which `verify` caught; now fixed).

Removed: automatic live deploys on every push to `main` .

Player impact: none yet. Players only benefit once you publish, because broken changes can no longer reach them silently.

Acceptance results

#	Criterion	Result	Evidence
A1	`verify` exits 0 on current code; non-zero with a broken	PASS	Local: exit 0 (63 tests, build), then exit 1 with one test deliberately broken ("1 failed, 62 passed"), then exit 0

#	Criterion	Result	Evidence
	test		once restored
A2	Push to `main` does not change the live site	PASS	Push of `2e4b240` started only *CI* (run 40274184197) and *Android build*; no live deploy run
A2	*Deploy* runs only after `verify` passes	PASS (by design) / NOT RUN (live)	`verify` is the step before deploy; not run because publication isn't approved
A3	A preview lets you play before publishing	PASS	Preview job: "Deployed fruit-market-staging", then "SMOKE OK … rules version 2" (version `adf6c304`)
A4	Owner can open the preview and finish a match using the brief	NOT RUN	Needs you (OWNER-BRIEF §1–2)
A5	`CLAUDE.md` under 120 lines; STATE about 500–600 words	PASS	72 lines; STATE about 350 words
—	Full CI (`verify:full`) on `2e4b240`	PASS	Checks job green in 14 min: lint, types, 63 tests, build, solo game, performance budget, online match with a leaver (0 leaks), quick match
—	Smoke script fails on a dead address	PASS	Local: "SMOKE FAILED … fetch failed", exit 1
—	Preview online rooms (Durable Objects) in a real browser	NOT RUN	The smoke test checks the page and API only. Covered by the optional online step in the playtest checklist

Deviation from the plan (needs your confirmation)

PLAN-001 said "a private Cloudflare preview URL; if it isn't free, local preview steps". Cloudflare's docs state that preview and version URLs **are not generated for Workers that implement a Durable Object**, and Fruit Market's server uses Durable Objects for rooms. So I used the next closest option: a second, separate Worker (`fruit-market-staging`). This is recorded as **DEC-006**.

- Cost: no new paid service. It runs within the Cloudflare plan you already use; usage is tiny. Your plan type is still unknown (DEC-005).
- "Private" means unlisted: anyone with the link can open it.
- To undo: delete the `fruit-market-staging` Worker in Cloudflare and remove the `preview` job.

Risks and costs

- Expected extra cost: RM0 (no new paid service). Not yet confirmed against your plan type.
- Until branch protection is on, a direct push can still skip CI, but it can no longer reach players without *Deploy*.
- If CI is red, the preview isn't updated; the live game is unaffected.

Decisions

ID	Decision	Approval status	Delivery
DEC-002	Package A scope = PLAN-001	Approved (owner, 2026-10-26)	Verified (A4 pending)
DEC-003	Live deploys only via manual *Deploy* after `verify`	Approved (owner, 2026-10-26, as part of PLAN-001)	Verified
DEC-006	Preview = separate staging Worker, because per-version URLs don't support Durable Objects	Awaiting confirmation	Implemented

Sources

- Cloudflare docs, Version URLs and Preview URLs (`developers.cloudflare.com/workers/versions-and-deployments/version-urls/` and `/preview-urls/`): not generated for Workers that implement a Durable Object. Retrieved 2026-10-26 via search.
- Wrangler config reference vs the Durable Objects environments page disagree on per-environment migrations; resolved empirically: the staging deploy with top-level migrations succeeded.
- CI run 40274184197 (checks and preview jobs green).

REP-003 · DEC-006 confirmed; PLAN-002 (Package B) proposed

Game	Fruit Market
Report date	2026-10-26 13:50 (Asia/Kuala_Lumpur)
Build	`main @ 63c3682` (live game still `c915e49`)
Type	Planning only. **No game implementation changes were made.**
Approval status	DEC-006 **approved** (owner, 2026-10-26: "Confirm DEC-006"). PLAN-002 **awaiting approval**

1 · DEC-006 confirmed

The preview stays a separate staging Worker (`fruit-market-staging`). Approval status: approved; delivery: verified (REP-002, A3).

2 · Problem Package B solves

Records exist, but they're kept by hand:

- Each decision's status is typed in two or three places (the register, every report that mentions it, STATE). Nothing checks that they agree. REP-002 already needed a manual word-count correction.
- Nothing stops an approval being recorded without your words as evidence, or history being edited.
- There's no player-facing changelog and no version tags, so "what changed in this release" can't be answered from the repo.
- The PDF export works (built in REP-001) but isn't part of any check.

Players never see this directly. It matters because it keeps approvals honest and makes releases traceable when there's money or player data involved later.

3 · PLAN-002 · Package B: canonical records, generated views, checks

Included scope

1. **One canonical source per record** (JSON, no new tools):
   - `docs/workflow/records/decisions.json`: every DEC with question, choice, rationale, impacts, owner, approval evidence (your exact words and date), delivery status, links, review trigger, supersession. A `history` list is only ever appended to.
   - `docs/workflow/records/PLAN-###.json`: one per package with scope, exclusions, acceptance, approval evidence, changes (Added / Changed / Fixed / Removed), checks, build, release state.
   - Reports (`reports/REP-###.md`) stay as hand-written narrative snapshots and are never rewritten.
2. **Generated views**, built by `npm run records` with a deterministic script:

    - `DECISIONS.md`: current decisions and pending approvals first, then the full history.
    - `CHANGELOG.md`: player-facing, grouped by release, with "unreleased" vs "published" state. It is seeded with one summary entry for everything before the workflow, taken from git history.
    - The "Pending approvals" block inside `STATE.md` (between markers; the rest stays hand-written).
    - PDFs: each report, plus the cumulative register (current decisions, then pending approvals, then all reports in order).
3. **Checks**, via `npm run records:check`, added to `npm run verify` so CI runs it. It fails when:
    - an ID is duplicated or skipped, or a report mentions a DEC, PLAN or REP that doesn't exist
    - a decision is "approved" without owner evidence, or a delivery status isn't one of the allowed values
    - a history entry was changed or removed compared with the last commit (append-only)
    - a generated view is out of date with its source
4. **Release tagging**: when you publish, the deploy records the release in the changelog and tags the commit (`v1.1.1` and so on), so every live build can be traced. This only runs when you press *Deploy*.
5. **Migration**: move DEC-001 to DEC-006, PLAN-001 and REP-001 to REP-003 into the new records without losing anything.

Excluded

- Game code, rules, server and hosting.
- Package C (devices, backups, capacity).
- A cross-game studio register (NOT APPLICABLE until there's a second game).
- Installing anything. PDF building stays local because it uses pandoc. If pandoc is missing, the script prints PDF EXPORT BLOCKED and keeps the last good PDFs; it isn't part of CI.

Roles: producer (record design), technical lead (scripts), QA (failure cases), DevOps (tagging). Self-review only; low risk.

Likely files: new `docs/workflow/records/*.json`, new `scripts/records.mjs`, new `CHANGELOG.md`, `docs/workflow/export-pdf.cjs` (reads the records), `package.json` (scripts), `.github/workflows/deploy-web.yml` (tag and changelog step), `CLAUDE.md` and `OWNER-BRIEF.md` (one line each).

Acceptance (observable)

- B1. `npm run records` rebuilds all views. Running it twice in a row changes nothing (no git diff in the Markdown outputs).
- B2. `records:check` fails, with a plain-English message, for each of these deliberate breaks (shown, then reverted):
    - a duplicate ID
    - an unknown DEC referenced in a report
    - "approved" without evidence
    - an edited history line
    - a stale `DECISIONS.md`

- B3. With pandoc made unavailable, the export prints PDF EXPORT BLOCKED, exits non-zero and leaves the previous PDFs untouched.
- B4. The cumulative PDF opens with current decisions and pending approvals, followed by REP-001 to REP-003 in order.
- B5. After migration, every decision, status and approval quote from REP-001 to REP-003 appears in the records.
- B6. `npm run verify` (and CI) runs `records:check` and stays green on the migrated records.
- B7. Tagging is tested on a throwaway branch: a dry run shows the tag and changelog entry it would create. The real tag happens only at your next publication.

Checks: B1–B7, plus `verify`, plus one CI run.

Cost and data: RM0. No player data. `npm run verify` gets about 1 second slower.

Risks:

- Record-keeping overhead: JSON is less friendly to edit by hand. Mitigation: Claude edits the records; you only read the generated Markdown and PDFs.
- The append-only check compares with the last commit, so a deliberate correction needs a new "correction" history entry rather than an edit. That is intended.

Rollback: revert the package commits. The hand-written `DECISIONS.md` and reports remain valid.

4 · Decisions

ID	Question	Recommendation	Status	Delivery
DEC-006	Preview = separate staging Worker	Keep	Approved (owner, 2026-10-26)	Verified
DEC-007	Package B scope = PLAN-002	As above	Awaiting approval	Not started
DEC-008	Canonical record format	JSON records plus Markdown narrative reports; Markdown, changelog and PDF views generated	Awaiting approval (part of PLAN-002)	Not started
DEC-009	Records check in `verify` and CI	Yes, so disagreements block merges once branch protection is on	Awaiting approval (part of PLAN-002)	Not started

Options considered for DEC-008:

- Keep hand-written Markdown only: free, but can't be validated reliably.
- YAML front matter: friendlier, but needs a parser dependency.
- A database: ruled out by the workflow guide.
- **JSON (recommended):** Node reads it without dependencies, and the check can be strict.

5 · Still open from earlier (not blocking PLAN-002)

- Which real phones you can test on (needed for Package C).
- Your Cloudflare plan (Free or Workers Paid) and any other paid accounts (DEC-005).
- Preview playtest (REP-002, A4) and branch protection (owner brief, section 5).

REP-004 · PLAN-002 (Package B) implemented and verified

Game	Fruit Market
Report date	2026-10-26 14:10 (Asia/Kuala_Lumpur)
Package	PLAN-002, approved by the owner 2026-10-26: "Approve PLAN-002 only, including its listed tests and documentation"
Build	`main @ a7fa577` (records), plus this report
Live game	**Unchanged** (still `c915e49`, version 1.1.0). No publication.
Delivery status	Verified (B1–B7 PASS)

What changed

Added

- `docs/workflow/records/`, the single source for every record:
  - `decisions.json`: all decisions, plus a history that can only grow
  - `PLAN-001.json` to `PLAN-003.json`
  - `releases.json`
- `npm run records`: rebuilds `DECISIONS.md`, `CHANGELOG.md`, the pending-approvals block in `STATE.md`, and every PDF.
- `npm run records:check` (now the first step of `verify`, so CI runs it). It rejects:
  - broken or duplicate IDs
  - references to things that don't exist
  - an approval without your words and a date
  - an edited history line or a rewritten approval
  - out-of-date views
- `CHANGELOG.md`, in plain player language. It is seeded with version 1.1.0 (what's live now).
- Release step in *Deploy LIVE game*: it refuses to run without release notes for the new version, and after a successful deploy it tags the commit (for example `v1.1.1`).

Changed

- The PDF export is all-or-nothing. On failure it prints PDF EXPORT BLOCKED and keeps the previous PDFs.
- CI fetches the full history so the records check can compare with the previous commit.

Removed

- The hand-edited `DECISIONS.md` (it is now generated).

Player impact: none directly. The release process is now traceable: every live build will have a version tag and changelog notes.

Acceptance results

#	Criterion	Result	Evidence
B1	Rebuilding twice changes nothing	PASS	Checksums of all three generated files identical after a second run
B2	Check fails on 5 deliberate breaks	PASS	Duplicate DEC-008; REP-003 citing a nonexistent DEC; DEC-001 "approved" with no evidence; history line 1 edited; stale register. Each exits 1 with a plain message. All reverted
B3	No pandoc → PDF EXPORT BLOCKED, old PDFs kept	PASS	Exit 1; all 4 PDFs byte-identical; no temporary files left
B4	Cumulative PDF order	PASS	Pending approvals, then current decisions, then REP-001, REP-002, REP-003
B5	Migration lost nothing	PASS	Every row of the old register matches. DEC-004 "In use" maps to the allowed value "implemented"; DEC-007 to DEC-009 changed only because of today's approval (in history)
B6	`verify` and CI run the check, green	PASS	CI run 40276699547: "RECORDS OK … history checked against 63c3682", full suite green, preview updated
B7	Tagging dry run	PASS	Throwaway branch: refuses while 1.1.0 is published; with a 1.1.1 candidate, "would tag v1.1.1" plus the notes. Branch deleted, no tag created

Known issue found (not part of this package)

The game screen sometimes doesn't appear after rejoining an online room. The automated tests caught it in 2 of the last 5 CI runs, with no game-code change between them:

- Run 1: after quick match, the second player's game didn't show within 15 s.
- Run 4 (a docs-only commit): after reloading the page mid-match, the game didn't come back within 20 s.

Every other run passed, including locally (more than 8 runs). Real players could hit this when their phone reloads the page or their connection drops.

This is pre-existing and intermittent; the cause isn't known yet. I recommend it as the next package, before Package C: reproduce it by repeating the reconnect test many times, find the cause, fix it, and prove the fix with repeated runs.

Risks and costs

- RM0. `verify` is about 1 second slower.
- The first live release under the new process needs "PREPARE RELEASE" first, because Deploy refuses without release notes. That is intended.
- Until the reconnect issue is fixed, CI can occasionally fail for reasons unrelated to the change being tested. Read red runs before assuming the latest change caused them.

Decisions

ID	Decision	Approval	Delivery
DEC-007	Package B scope = PLAN-002	Approved (owner, 2026-10-26)	Verified
DEC-008	JSON records plus narrative reports; generated views	Approved (owner, 2026-10-26)	Verified
DEC-009	Records check in `verify` and CI	Approved (owner, 2026-10-26)	Verified

Still pending your decision: DEC-001 (adopt the workflow), DEC-004 (PDF tooling), DEC-005 (budget assumption) and PLAN-003 (outline). See the register's "Pending approvals".

REP-005 · PLAN-004 proposed: fix the rejoin problem in online play

Game	Fruit Market
Report date	2026-10-26 14:20 (Asia/Kuala_Lumpur)
Build	`main @ 935a287` (live game still 1.1.0, `c915e49`)
Type	Planning only. **No game implementation changes were made.** Read-only code inspection only.
Approval status	PLAN-004 **awaiting approval** (DEC-010, DEC-011)

1 · The problem

Sometimes, after a player rejoins an online room, the game screen doesn't come back:

CI run	Situation	Waited	Result
Run 1 (`da2eea9`)	Second player joins via Quick match	15 s	Game screen never appeared
Run 4 (`63c3682`, docs only)	Player A reloads the page mid-match	20 s	Game screen never appeared
Runs 2, 3 and 5, plus 8+ local runs	Same steps	—	Passed (usually about 1 s)

There was no game-code change between the failing and passing runs, so the problem is timing-dependent. Real players can trigger it: phones reload pages when memory is low, and mobile connections drop. The tests saved **no evidence** when it happened: the local server's log was discarded, and the online test takes no screenshot on failure.

2 · What the code shows (read-only)

How rejoining works today:

1. The phone keeps a seat token per room.
2. On reload, the online screen sees `?room=CODE` plus the token and reconnects.
3. The server's `hello` matches the token to the seat and sends the room state.
4. The game screen appears once the state says the match has started.

Hypotheses, none confirmed yet:

#	Hypothesis	Where	Fits run 1	Fits run 4	Evidence so far
H1	In a quick-match lobby, a seat is deleted when its connection closes. If the waiting player's connection	`server/room.ts closed()`	Yes	No	Plausible from code

#	Hypothesis	Where	Fits run 1	Fits run 4	Evidence so far
	blips just as the stranger joins, the match never starts				
H2	The old connection's "closed" event reaches the server *after* the new `hello`, wrongly marking the seat offline or freeing it	`server/room.ts` `closed()` / `hello()`	Yes	Yes	Plausible; order depends on server timing
H3	The server answers the rejoin with a non-fatal error and the screen stays on "Connecting…" with no retry	`server/room.ts`, `src/ui/Online.tsx`	Maybe	Yes	No error is shown or logged in the test
H4	The client drops or ignores the first room state after reconnecting (version check or listener timing)	`src/net/client.ts`, `useGameController.ts`	Maybe	Yes	Code looks right, but it isn't tested under slow timing
H5	Only the test environment: an overloaded CI machine is slower than the timeout	Test scripts	Yes	Yes	Normal time is about 1 s, so 15–20 s would be extreme. Not assumed without data

Ruled out: a seat being lost when the server sleeps. Seats are stored on each connection in a way that survives the server sleeping (`serializeAttachment`).

3 · PLAN-004 · Reconnect fix (evidence first)

Player problem: a player who reloads or loses connection can be stuck on a waiting screen instead of returning to their match.

Included scope (small steps)

1. **Evidence.** When the online or quick-match test fails, it saves:
    - both players' screens, page text and browser console
    - the room messages each player received
    - the local server log, kept as a CI artifact for 7 days
2. **Reproduce.** A new stress test, `npm run test:reconnect`, repeats three kinds of cycle and times how long the game screen takes to return:
    - reload mid-match
    - a connection drop (the browser goes offline, then online)
    - quick-match pairing

    It runs normally and with the CPU slowed 4× (like a cheap phone). It can be run locally and from a manual CI workflow (DEC-011). I'll run it in batches (target 300 cycles) and record the failure rate.
3. **Isolate.** Compare failing and passing cycles to confirm or reject H1–H5.

4. **Fix the measured cause** with the smallest change, plus a regression test that fails before the fix and passes after.
    - If the cause is in seat or token handling (who owns a seat), a fresh, separate review checks the change, because that's where security matters.
    - If the cause is only test timing (H5), the fix goes in the tests, but only with stress-test data showing real rejoins work.
5. **Prove.** The stress test and CI must stay green repeatedly (acceptance below).

Excluded

- Rules, balance, `RULES_VERSION` and saves.
- Message-format changes that old app versions can't understand (if one is needed, I stop and replan).
- Publishing (that needs "PREPARE RELEASE" and your Deploy).
- Package C.
- Making tests "pass" by just waiting longer or retrying.

Roles: network/backend engineer and QA (lead), technical lead, plus security review if seat or token logic changes.

Likely files:

- `scripts/e2e-online.cjs`, `scripts/e2e-quick.cjs`, `scripts/e2e.mjs` (diagnostics, server log)
- new `scripts/e2e-reconnect.cjs`
- new `.github/workflows/reconnect-stress.yml` (manual)
- `package.json`
- then, depending on the cause: `server/room.ts` or `src/net/client.ts` or `src/ui/Online.tsx`, plus a test in `tests/room.test.ts`

Acceptance (observable)

- R1. `npm run test:reconnect` prints time-to-game for every cycle, the failures, and the 95th percentile.
- R2. The root cause is named with evidence from a failing cycle (a log or message excerpt). If it isn't reproduced after at least 300 cycles, I report **NOT REPRODUCED** with the conditions tried, and stop rather than guess.
- R3. If reproduced: the failure rate before the fix is recorded, and after the fix at least 100 slowed-down cycles have 0 failures.
- R4. A deliberately forced failure in the online test produces the diagnostics files and the server log artifact.
- R5. A regression test fails on the old code and passes on the fixed code.
- R6. `verify:full` is green on 3 consecutive CI runs; the leak check stays at 0; all unit tests stay green; no rules or save changes.

Checks: R1–R6, `verify`, the manual stress workflow, and CI.

Cost and data: RM0. GitHub Actions is free for public repositories. Diagnostics contain only test players and expire after 7 days.

Risks

- It may not reproduce outside CI. Mitigation: the slowed-down mode and the CI workflow, and a hard stop at 300 cycles with a report.
- A fix in seat handling could affect reconnect security. Mitigation: a regression test, the existing room tests, and a separate review.
- Time: likely 1–2 sessions. I'll report at the end of step 3 if the cause turns out larger than a small fix.

Rollback: revert the package commits. Players are unaffected until a release is published.

How you'll see the result: once the fix is on the preview, reload the page mid-match on your phone, or switch to airplane mode for 5 seconds and back. The match should return within a few seconds.

4 · Decisions

ID	Question	Recommendation	Status
DEC-010	How to fix the rejoin problem	Evidence and reproduction first, then fix only the measured cause (PLAN-004)	Awaiting approval
DEC-011	Run the stress test on every push?	No: on demand (manual workflow and local runs)	Awaiting approval

5 · Still open

- Pending decisions from earlier: DEC-001, DEC-004, DEC-005, and PLAN-003 (outline only).
- Your preview playtest, branch protection, your phones and your Cloudflare plan.

REP-006 · PLAN-004 done: the online "stuck on Connecting..." bug is found and fixed

Game	Fruit Market
Report date	2026-10-26 15:25 (Asia/Kuala_Lumpur)
Package	PLAN-004, approved by the owner 2026-10-26: "Approve PLAN-004 only, including its listed tests and documentation"
Build	`main @ d7e5d24` (fix in `9d9e911`)
Live game	**Unchanged** (1.1.0). The fix is on the preview; players get it only after "PREPARE RELEASE" and your Deploy
Delivery status	Verified (R1–R6 PASS)

The bug, in plain words

After a player reloaded the page or rejoined an online match, the game sometimes stayed on **"Connecting to room ..."**, even though the match was there and the server had already sent it.

Why: the screen started listening for room updates a moment *after* it first appeared. The connection was already open by then. If the server answered in that gap, the answer was missed, and the screen waited for a next message that might never come (for example, when everyone was waiting on that player).

The gap is tiny on a fast computer, so it usually worked. On a slow phone, or when the app is in the background (browsers pause drawing for hidden tabs), the gap is long and the bug happens often. The game screen had the same weakness for updates arriving just as it opened.

Fix: two small changes in the app (`src/ui/Online.tsx`, `src/ui/game/useGameController.ts`). Right after starting to listen, each screen checks what has already arrived.

- No server change, no message-format change, no rules or save change.
- Old and new app versions stay compatible.

Evidence

Check	Result
Normal timing, old code (local)	270 cycles (reloads, network drops, quick matches; normal and 4× CPU slowdown), 0 failed. The bug is too rare to see this way
Background-tab mode, **old code**	**20 of 20 reloads failed**; every one stuck on "Connecting to room" with the full game already received (`room v2 status=playing`)
Background-tab mode, **fixed code** (local)	65 cycles, 0 failed

Check	Result
Background tab plus 4× slowdown, **fixed code** (local)	100 cycles, 0 failed (reload p95 0.59 s, drop p95 0.63 s, quick match p95 1.3 s)
Same on a CI machine (where the original failures happened)	110 cycles, 0 failed (Reconnect stress run 40283316566)

Acceptance results

#	Criterion	Result	Evidence
R1	Stress test with per-cycle timings	PASS	`npm run test:reconnect` ; modes: slowdown, background tab, preview target
R2	Root cause named with evidence	PASS	Lost first update after (re)joining; see the old-code evidence above
R3	Before rate recorded; at least 100 slowed cycles with 0 failures after	PASS	20/20 failed before; 100/100 passed after (plus 175 more)
R4	Forced failure leaves evidence	PASS	Page text, room messages, console, both screenshots and the server log were saved
R5	Regression test fails before, passes after	PASS	New suite step "rejoin from a background tab"
R6	3 consecutive green CI runs; leak check 0; unit tests green; no rule or save changes	PASS	CI runs 7 (push), 8 and 9 (manual) on `d7e5d24` : all green, each with leaks 0 and 63 unit tests; no rules or save files changed

Side observations (no action taken)

- **The local test server never finishes closing a socket when the page closes it.** Cloudflare's docs say production replies automatically for this project's settings, and the real app never waits on a socket it closed itself, so no player impact was found. If it ever matters, the documented safe fix is one line in the server's close handler.
- **Playwright's offline mode doesn't break open connections**, so the stress test cuts them with a small local relay instead. A "silent" dead connection (no signal at all) isn't simulated yet. That would be part of Package C.

What changed

Added

- `npm run test:reconnect` and a manual *Reconnect stress* workflow (local or preview, optional slowdown and background-tab mode).
- Failure evidence for all online browser tests.
- The regression step in the browser suite.

Fixed (player-facing, on the preview, not yet live)

- After reloading or rejoining an online match, the game could stay stuck on "Connecting to room...", especially on slower phones or when the app was in the background.

Risks and costs

- RM0.
- The browser suite is about 1 minute longer (the regression step).
- The fix affects all online screens. Mitigation: the full suite, including the secret-price leak check, stayed green, and 545 stress cycles ran in total.
- Remaining NOT RUN: a real phone in a real background tab. Your preview playtest covers that (see below).

How to try it (preview)

1. Open https://fruit-market-staging.murshidi-hashim.workers.dev on two phones (or one phone and a computer).
2. Start an online game (Play a friend → Create room → open the link on the other device).
3. During the match, on one phone: switch to another app for 10 seconds, come back, then pull down to reload.
4. Expected: the match comes back within a few seconds both times, never stuck on "Connecting to room...".